

KENNETH CALHOUN

Senior Backend Engineer | Scalable Systems & API Architecture | Cloud, Python, Databases

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SUMMARY

Senior Backend Engineer with 10+ years of experience designing and scaling cloud-native platforms across fintech, eCommerce, and real-time systems. Built high-performance APIs, payment platforms, and distributed services that process millions of transactions and power global applications. Currently leading backend architecture and engineering teams to deliver secure, reliable, and low-latency systems on AWS, Kubernetes, and PostgreSQL. Seeking to apply technical leadership and deep expertise in backend architecture to fintech and cloud innovation projects, while mentoring engineers and driving product growth at scale.

SKILLS

- **Languages & Frameworks:** Python (Django, Flask, FastAPI), Go, SQL
 - **Architecture:** Microservices, REST, gRPC, Serverless
 - **Databases:** PostgreSQL, MySQL, Redis, Metabase
 - **Infrastructure & DevOps:** AWS (Lambda, EC2, RDS, S3, API Gateway), Docker, Kubernetes, Terraform, CI/CD pipelines
 - **Messaging & Data:** Celery, RabbitMQ, Kafka, Prometheus, Grafana, HAProxy, Postgres
 - **Security & Compliance:** RBAC, Encryption, TLS, SHA, JWT
 - **Other:** Payment Systems, High-frequency trading systems, Platforms, Financial Credit Systems
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WORK EXPERIENCE

Lead Backend Engineer | Python, Cloud & API Architecture

A-CX, San Jose, CA

January 2022 – June 2025

- Orchestrated backend development of a Buy Now, Pay Later (BNPL) platform using Python, Django, PostgreSQL, and AWS, supporting millions of dollars in monthly transaction volume.
- Integrated multiple payment gateways (Stripe, PayPal, local banks) with real-time risk assessment, accounting, and fraud detection modules.
- Engineered a microservices-based backend capable of handling 10K+ concurrent daily transactions, with end-to-end encryption and audit logging to meet financial compliance requirements.

- Enhanced database queries and APIs, reducing checkout response times by 50%+, directly improving the user payment experience.
- Supervised a team of 8 engineers in delivering a fully managed PostgreSQL cloud service, operating hundreds of clusters (100TB+ data) with 99.99% uptime.
- Devised a Kubernetes operator with configurable master–replica setups, daily S3 backups, and Point-in-Time Recovery, cutting user-reported issues by 50%.
- Automated cluster provisioning, monitoring, and billing with Prometheus, Grafana, and Kubernetes orchestration, reducing manual operations by 40% and improving reliability at scale.

Senior Software Engineer – Backend Systems | Python, Microservices, AWS

Microsoft, Redmond, WA

January 2019 – December 2021

- Maintained Python-based highly available backend services for the Microsoft App Store homepage, ensuring low latency under peak global traffic.
- Deployed event-driven microservices to streamline app catalog updates and content delivery, reducing sync times from hours to minutes.
- Collaborated with data science teams to embed AI-driven recommendation models into the personalization pipeline, improving app discovery relevance and driving higher user engagement.
- Advanced observability and reliability with AWS CloudWatch, Prometheus, and custom dashboards, cutting mean time to resolution (MTTR) for incidents by 40%

Senior Backend Engineer – Product Platform Optimization

Zenesys, Frisco, TX

January 2017 – December 2018

- Constructed backend trading services that processed over \$50M in monthly crypto volume, maintaining low execution latency.
- Authored APIs for order execution, risk checks, and market data, enabling seamless integration with major crypto exchanges.
- Migrated systems to AWS microservices to improve scalability and fault tolerance, achieving 99.99% uptime even during peak trading hours.
- Introduced real-time monitoring and automated alerting, reducing incident response time by 40% and preventing costly downtime.
- Partnered with quant teams to incorporate AI-driven pricing and prediction models into the trading pipeline, improving spread optimization and reducing slippage.

Python Backend Developer | Django, PostgreSQL, REST APIs

Zenesys, Frisco, TX

July 2013 – December 2016

- Coded backend services for Roco, a UK-based fashion eCommerce platform, using Django and Optimizely Commerce to support 50K+ products, multiple vendors, and high seasonal traffic.
- Devised smart search, product filtering, and REST APIs for order management, vendor dashboards, and Firebase-based real-time notifications, reducing search latency by 40% and enabling seamless checkout with faster delivery updates.
- Integrated Razorpay payment gateway with UPI, wallets, and cards, ensuring secure multi-option payments and compliance with invoicing/tax regulations.
- Deployed and scaled services on AWS cloud with CI/CD pipelines, achieving 99.9% uptime during peak sales campaigns and improving deployment speed by 35%.

Python Developer Intern | Flask, SQL, Unit Testing

Zenesys, Frisco, TX

July 2012 – June 2013

- Assisted in developing RESTful APIs with Flask, implemented SQL queries for data retrieval, and created unit tests to improve backend service reliability.
 - Contributed to QA and continuous integration processes, performing code reviews and automated testing to improve code quality and maintainability.
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EDUCATION**Bachelor of Science in Computer Science**

Finlandia University, Hancock, MI

April 2008 – May 2012